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Scientific Computing

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2.6: convergence of implicit Euler

Outline

Lecture 2.6: convergence of implicit Euler

- Theorem: global error bound
- Error recurrence with factor $1/(1 - hL)$
- Estimate of $(1 - hL)^{-N}$ and final bound
- Q & A

Series: 2.1 Euler schemes. 2.2 errors. 2.3 geometry. 2.4 explicit convergence. 2.5 Grönwall. 2.6 implicit convergence.

Theorem: convergence of implicit Euler

Theorem 2.1 (Convergence of implicit Euler: $\|e_n\| \le \frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h + O(h^2)$ for $hL \le \frac{1}{2}$)

In the setting of the explicit case (f Lipschitz with constant L, $u \in C^2$, $\|u''\| \le M$) let

$$y_{n+1} = y_n + h\, f(t_{n+1}, y_{n+1}), \qquad t_n = \alpha + nh, \qquad hL \le \tfrac{1}{2}.$$

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Then

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Remark 2.2 (Why $hL < 1$: the implicit Euler equation has a unique solution (contraction))

$hL < 1$ makes the map $z \mapsto y_n + hf(t_{n+1}, z)$ a contraction, so y_{n+1} exists, is unique and fixed-point iteration converges.

Proof of implicit Euler convergence, step 1: LTE

Proof of implicit Euler convergence, step 1: local error $\|\tau_{n+1}\| \leq Mh^2/2$.

- Taylor expansion of $u(t_n)$ around t_{n+1}:

$$u(t_{n+1}) = u(t_n) + h\, f(t_{n+1}, u(t_{n+1})) + \tau_{n+1}, \qquad \tau_{n+1} = -\frac{h^2}{2} u''(\xi_{n+1}).$$

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- Hence $\|\tau_{n+1}\| \le \dfrac{Mh^2}{2} = O(h^2)$.

□

Proof of implicit Euler convergence, step 2: error recurrence

Proof of implicit Euler convergence, step 2: recurrence with factor $C = 1/(1 - hL)$.

- Subtract the implicit Euler step from the Taylor identity and take norms; the Lipschitz term now involves e_{n+1}:

$$\|e_{n+1}\| \leq \|e_n\| + hL\|e_{n+1}\| + \|\tau_{n+1}\|.$$

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- Move the e_{n+1} term to the left, $(1 - hL)\|e_{n+1}\| \leq \|e_n\| + \|\tau_{n+1}\|$, and divide by $1 - hL > 0$. With $C = \dfrac{1}{1 - hL}$:

$$\|e_{n+1}\| \leq C\|e_n\| + C\|\tau_{n+1}\|.$$

□

Proof of implicit Euler convergence, step 3: unfolding

Proof of implicit Euler convergence, step 3: unfolding to $\|e_N\| \le \frac{Mh}{2L}(C^N - 1)$.

- Unfold with $e_0 = 0$:

$$\|e_N\| \le \sum_{k=0}^{N-1} C^{N-k} \|\tau_{k+1}\| \le \max_k \|\tau_{k+1}\| \sum_{j=1}^{N} C^j$$

$$= \max_k \|\tau_{k+1}\| \, \frac{C(C^N - 1)}{C - 1}.$$

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- $C - 1 = \dfrac{hL}{1 - hL}$, so $\dfrac{C}{C-1} = \dfrac{1}{hL}$. With $\|\tau_{k+1}\| \le \frac{Mh^2}{2}$:

$$\|e_N\| \le \frac{Mh}{2L}\left(C^N - 1\right).$$

□

Lemma: bound on the amplification factor

Lemma 2.3 (Estimate $(1-hL)^{-N} \le e^{L(t_N-\alpha)}e^{Kh}$ for $hL \le \frac{1}{2}$)

If $0 < hL \le \frac{1}{2}$ *and* $t_N - \alpha = Nh$, *then*

$$\left(\frac{1}{1-hL}\right)^N \le e^{L(t_N-\alpha)}\, e^{Kh}, \qquad K = L^2(t_N-\alpha).$$

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Proof of the estimate of $(1-hL)^{-N}$.

- $\ln C^N = -N\ln(1-hL) = N\left(hL + \frac{(hL)^2}{2} + \frac{(hL)^3}{3} + \cdots\right)$.

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- For $x = hL \le \frac{1}{2}$: $\frac{x^2}{2} + \frac{x^3}{3} + \cdots \le x^2$.
- So $\ln C^N \le N(hL + h^2L^2) = L(t_N-\alpha) + L^2(t_N-\alpha)\,h$, using $Nh = t_N - \alpha$.

□

Proof of implicit Euler convergence, step 4: final bound

Proof of implicit Euler convergence, step 4: final bound and first order.

- Insert the lemma into $\|e_N\| \leq \frac{Mh}{2L}(C^N - 1)$:

$$\|e_N\| \leq \frac{Mh}{2L}\left(e^{L(t_N-\alpha)}e^{Kh} - 1\right).$$

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- Write $e^{Kh} - 1 = O(h)$:

$$e^{L(t_N-\alpha)}e^{Kh} - 1 = \left(e^{L(t_N-\alpha)} - 1\right) + O(h).$$

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- Therefore

$$\|e_N\| \le \frac{M}{2L}\Big(e^{L(t_N-\alpha)} - 1\Big)h + O(h^2) = O(h).$$

- Implicit Euler is consistent (LTE $O(h^2)$) and converges with order 1.



References

- **Implicit Euler scheme.** J. Beurich, *Differences of solutions of implicit Euler schemes with accretive operators on Banach spaces*, `arXiv:2408.13524` (ref.).

Q & A